Learning Management System (LMS) - Product Requirements Document (PRD)

1. Product Overview

The Learning Management System (LMS) is a platform designed to organize and deliver structured learning using YouTube-based video lessons.

The system enables instructors to create courses, students to enroll and watch lessons sequentially, and administrators to manage the platform.

The LMS tracks learning progress, controls lesson order, and ensures structured video-based learning.

2. Goals & Objectives

• Provide structured learning using video lessons.

• Track student progress and completion.

• Allow course creation and organization into sections and lessons.

• Ensure lessons are watched in order.

• Provide simple, minimalist UI for learning.

3. Target Users

Students – enroll in courses and watch lessons.

Instructors – organize course content.

Administrators – manage platform content and users.

4. Key Features

• User Authentication (Register/Login)

• Course Listing Page

• Course Details Page

• Structured Lesson Navigation

• YouTube Video Embedding

• Lesson Progress Tracking

• Resume from Last Watched Position

• Sequential Lesson Unlocking

• Student Dashboard & Progress Overview

5. User Flow (Student Perspective)

1. Student registers or logs in.
2. Student views course list.
3. Student opens course details.
4. Student enrolls in course.
5. Student opens learning page.
6. Video player loads lesson video from YouTube.
7. Lesson progress is tracked.
8. Student moves to next lesson.

6. System Architecture

Frontend: Next.js + TailwindCSS (Minimalist UI).
Backend: Node.js with Express REST API.
Database: MySQL (Aiven).
Hosting: Backend on Render, Frontend on Vercel.

Videos are stored on YouTube. The LMS only stores video URLs and metadata.

7. Core Modules

Authentication Module – login, register, JWT authentication.
Course Module – course title, description, thumbnail.
Section Module – organizes lessons into groups.
Video Module – stores lesson metadata and YouTube links.
Enrollment Module – tracks course enrollment.
Progress Module – tracks video completion and watch time.

8. Database Entities

Users
Courses (Subjects)
Sections
Videos
Enrollments
Video Progress
Refresh Tokens

The database stores metadata only, not video files.

9. API Overview

Authentication APIs:

POST /api/auth/register

POST /api/auth/login

POST /api/auth/refresh

POST /api/auth/logout

Course APIs:

GET /api/subjects

GET /api/subjects/{id}

GET /api/subjects/{id}/tree

Video APIs:

GET /api/videos/{id}

Progress APIs:

GET /api/progress/subjects/{id}

POST /api/progress/videos/{id}

10. Security

• JWT Authentication
• Access Token (15 minutes)
• Refresh Token (30 days)
• HTTP-only cookies for refresh tokens
• Secure API authorization
• CORS configuration for frontend domain

11. Video Playback

Videos are embedded using YouTube IFrame Player API.

The system tracks playback progress and updates database periodically.

Completed videos unlock the next lesson.

12. Deployment

Frontend deployed on Vercel.

Backend deployed on Render.

MySQL database hosted on Aiven.

Environment variables store configuration such as database credentials and API URLs.

13. Success Metrics

- Number of enrolled users
- Lesson completion rate
- Average course progress
- Daily active learners
- System uptime and performance